Speaker Notes

Contents

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Slides: 64 · **Duration**: 60 min · **Venue**: Brno (community event)

1 1. Building AI Systems

On screen:

> *Brno tech community · 2026-09-16*
>
> *Reliability is engineered, not prompted.*
>
> *Ondrej (Ondra) Krajicek me@ondrejkrajicek.com · linkedin.com/in/OndrejKrajicek Built at render time*

If you write code, you should leave with something to build or refuse to build on Monday. [speculative] If you do not write code but you have to judge what these systems produce, you should leave with a question you can ask that gets a real answer. [speculative] And if you decide budgets, you should leave with the failure modes and the cost shape written down. [speculative] Every load-bearing claim tonight carries a number, a mechanism or a named source, because none of you should buy an adjective from a stranger. [speculative]

2 2. Reliability is engineered, not prompted

On screen:

> *The claim*
>
> *Two mechanisms, and neither is a prompt. [inference]*
>
> - ***External grounding*** *— a check that can still fail the work when the generator is wrong. A model approving its own output cannot: its errors correlate with the ones that produced the answer — the shared blind spot. [V3]*
> - ***Bounded autonomy*** *— a prompt is advisory: tool output can override it. A credential or gateway rule runs after the model and refuses the emitted call, so deny irreversible actions there. [V32] [inference]*
>
> *[V3] Loop Engineering and Self-Verification — vault report (2026). [V32] Harness Engineering — Tool Permissions — vault report (2026).*

Here is the whole argument, so you know what you came to disagree with. [speculative] First mechanism: external grounding. A check tells you something only if it can fail the work when the generator is wrong, and a model asked to approve its own output cannot — it shares every blind spot that produced the answer. [V3] So the load-bearing variable is not iteration count; it is whether the check can come back negative. [V3] Second mechanism: bounded autonomy. A prompt instruction is advisory — for an agent reading tool output, almost any turn brings text that overrides it. [inference] A credential scope or an API gateway rule is evaluated after the model, so it refuses the emitted call whatever the model produced. [inference] Deny irreversible actions there, not in the prompt. [V32] If by the end you think these are the wrong two variables, come and tell me which ones are.

3 3. "Done. I fixed it and verified everything passes."

On screen:

> *Opening*
>
> *Every seat in this room has read a sentence like that and found out otherwise. [speculative]*
>
> *1*
>
> *If you write code*
>
> *The assistant reports the fix, the suite is green, the defect ships. [speculative]*
>
> *2*
>
> *If you judge output*
>
> *A confident answer, a real-looking citation, a policy that was superseded. [speculative]*
>
> *3*
>
> *If you fund it*
>
> *The demo worked; the pilot did not; nobody can say which step failed. [speculative]*

You have all seen a system announce that it finished, in a completion message so specific and so calm that disagreeing looks unreasonable — and then it had not finished. [speculative] Each column is the version of that you have lived. [speculative] For the rest of the hour I will show why that shape happens and what specifically removes it. [speculative]

4 4. An agent auditing its own work caught 0 of 34 real violations

On screen:

> *Opening*
>
> *A false completion, counted: the agent reviewed its own output, stated high confidence, and was wrong on all 34 cases. [V1]*
>
> *0 of 34*
>
> *Real violations caught by the agent auditing its own work [V1]*
>
> *34 of 34*
>
> *The same violations caught by a deterministic check [V1]*
>
> *The agent rated its own confidence 90–100 on all 34 it missed. [V1] Confidence that does not move when the code is wrong cannot gate a merge. [inference]*
>
> *[V1] Loop Engineering and Self-Verification Loops — Practitioner Synthesis — vault research report (2026).*

Here is the false-completion report with a number attached. [speculative] An agent auditing its own work caught zero of thirty-four real violations, and it stated ninety to one hundred confidence while missing every one of them. [V1] A deterministic check, run over the same artifacts, caught all thirty-four. [V1] Not a better model — a program. [inference] Now look at what that confidence number did across the thirty-four cases: nothing. [inference] It did not move when the work was wrong, so it cannot tell you the work is wrong. [inference] I want to be careful about how far that generalises, because it is one measurement and not a theorem — where a model has an external signal to condition on, a stated confidence can carry real information. [speculative] Here it had none, and it read like certainty anyway. [inference]

5 5. Roadmap — vocabulary, the negative result, the external check, the gate, the bill

On screen:

> *Roadmap*
>
> 1. *The thing you already run: harness, agent, workflow, and the vocabulary for the hour*

> *2. Why a generator cannot be its own oracle — and four reasons that is not the whole story*
>
> *3. What an external check buys, what it costs, and where it goes blind*
>
> *4. Bounded autonomy — deny at the API gate, not in the prompt*
>
> *5. Cost, and one decision per seat in this room*
>
> *By the end you can answer two questions about a system you already run: which of your checks can fail when the model is wrong — a compiler, or a second prompt? And what may this agent do before a person approves it?*

Five parts, about ten minutes each. Part one names the machinery behind the tools, because plenty of people run one every day without using the word. Part two is the negative result at the centre of the talk, and then four serious reasons it is narrower than the slogan version. Part three is the positive half, with the bill and the blind spots attached. Part four I will argue is a permissions problem before it is an AI problem. The two questions on the slide are the take-home; the theory exists to earn them.

6 6. We cover the check around the model, and skip the model

On screen:

> *Scope*
>
> *The check around the model is where the gain was measured: structural code compliance 56.8% → 98.6%, credited to an external verifier, not a better model. [V4] [V5]*
>
> ***Covered***
>
> *• Where does the check sit, and can it fail when the model is wrong?*
>
> *• What may an agent do before a person says yes?*
>
> *• What does the check bill, in tokens and in review capacity?*
>
> ***Not covered***
>
> *• Prompt technique and model internals*
>
> *• Which model is best this month — not what bought the gain above [V5]*
>
> *• CI, code review, testing — assumed*
>
> *• Retrieval implementation and knowledge graphs*
>
> *[V4] [V5] Loop Engineering and Self-Verification — vault report.*

A fence around the hour, and one line on why the model layer sits outside it. [speculative] Not because it is unimportant. In the study I build the positive half of this talk on, the gain was not bought with a better model: structural code compliance went from fifty-six point eight to ninety-eight point six percent, credited to closing the loop against an external verifier, and swapping the backbone did not move it. [V4] [V5] One study, one domain, and it is the number I am asking you to act on. [inference] Retrieval implementation deserves its own hour. [speculative]

7 7. Takeaways — move the check outside the model, and authority out of the prompt

On screen:

> *Where this goes*
>
> *1*
>
> *Externalise the check*
>
> *Structural code compliance 56.8% → 98.6% (structural design — my own field). The authors credit the external verifier, not a better model. [V4] [V5]*
>
> *2*
>
> *Can it fail when the generator is wrong?*
>
> *Self-rechecking is not that check. [V3] [inference]*
>
> *3*
>
> *Name the check*
>
> *An exit code, a schema, a passing test. 70% of measured loops verify at the deterministic and rule levels. [V18]*
>
> *4*
>
> *Gate irreversible actions*
>
> *Deploys, money movement, force-push — on a deterministic deny gate, never a model. [V20]*

One: move the truth-determining computation out of the model — the study behind my headline number says reliability came not from a more capable model but from closing the loop against an external verifier. [V5] Two: a check counts only if it can come back negative when the generator is wrong. [V3] Running the same model over its own answer again does not change that. [inference] Three: before you trust a decision, name what actually checked it — an exit code, a schema, a passing test, a rubric score. Seventy percent of the loops measured verify at the deterministic and rule levels — an exit code, a schema —

rather than with a model. [V18] [inference] I want that to land as a good sign rather than as a limitation. [speculative] Four: add authority last — irreversible actions, meaning production deploys, money movement and force-push, belong on a deterministic deny gate, never on a probabilistic reviewer. [V20] If you disagree early, spend the hour building the rebuttal.

8 8. Models drafted this deck; I read every slide and rewrote what I disagreed with

On screen:

> *Disclosure*
>
> - ***Drafting*** *— eight persona sweeps, merged*
> - ***Critics*** *— seven models, nine scripts*
> - ***Quotes*** *— a script matches each quotation byte-for-byte to its note; a mismatch fails the build [inference]*
> - ***External check*** *— me, every slide*
>
> *My notes name the failure I guard against: "models agreeing with models about a memory curated by models" [V48].*
>
> *The claim is comparative, not prohibitive: a check able to fail the work carries information that fluent prose does not. Mine is the last row: a person. [inference]*
>
> *[V48] 2026-07-17 Digest.*

Eight model personas researched this over my own vault, a model merged their output, seven model critics read the merge, and nine deterministic checks ran over it. [speculative] My notes name what that arrangement invites — models agreeing with models about a memory curated by models. [V48] The scripts are the cheap half: a build script matches every quotation from my own notes byte-for-byte against the note it names, and a mismatch fails the build. [inference] The expensive half is that I read every slide and rewrote what I disagreed with — a person's time, and it does not scale. [speculative] Do not hear this as do not trust generated work. My claim is comparative: a check you can point at beats prose that reads well. Mine is a script and a person. [inference]

9 9. The thing you already run

On screen:

> *Part I*
>
> *Naming the machinery before arguing about it.*

Part one, and it is the shortest. [speculative] I am not going to define what a prompt is or explain what a large language model does; if you write code, most of you use these tools daily and that slide would insult you, and if you do not, nothing tonight turns on the model's internals — it turns on what sits around the model. [speculative] What I do want is four shared words, because this room comes from different companies and different stacks, and I have watched conversations about agents fail purely on vocabulary — two people using the word verifier for two different things and agreeing loudly while disagreeing completely. [speculative] So: harness, workflow, agent, oracle. [speculative] Four words, about eight minutes, and then we can argue about substance instead of terms. [speculative]

10 10. You already operate a harness

On screen:

> *Part I · Harness*
>
> *A harness is "everything in an AI agent except the model itself" — system prompts, tools, Model Context Protocol servers, sandboxes, persistent memory, custom linters, structural tests. [V21]*
>
> - *The coding assistants most engineers now open daily are harnesses; the model is one component inside them. [inference]*
> - *Karpathy's picture: "the LLM is the CPU and its context window is the RAM" [V50] — the harness is the operating system above both. [inference]*
> - *So the question "should we build an agent?" is usually really "whose harness are we adopting?" [speculative]*
>
> *[V21] Harness Engineering · [V50] Context Engineering — vault concept notes.*

Start with the word the other three definitions hang off. [speculative] A harness is everything in an AI agent except the model itself: system prompts, tools, the MCP servers that hand it those tools, sandboxes, persistent memory, custom linters, structural tests. [V21] If you write code, the assistant most of you opened this morning is a harness with a model inside it, rather than a model with a chat window. [inference] If you do not, the systems you are asked to judge or fund have the same shape: a model wrapped in code that assembles its context and enumerates the tools it may call. [inference] The picture that works across a mixed room is Karpathy's: the model is the CPU, the context window is the RAM, and

context engineering is memory management. [V50] The harness is the operating system above both — the code that loads context, dispatches tool calls, and checks permissions. [inference] Which reframes the buying question you are actually facing: not which model, but whose harness. [speculative]

11 11. The harness moved Terminal-Bench 2.1 scores about 15× further than the model — tune the harness first

On screen:

> *Part I · Harness*
>
> *0.3 pts*
>
> *Model swapped, harness held. Artificial Analysis: Opus 4.8, Fable 5 (Anthropic) 84.6%; GPT-5.5 84.3% [V56]*
>
> *4.3–5.2 pts*
>
> *Harness swapped, model held. tbench.ai board: GPT-5.5 83.4% Codex CLI → 78.2% Terminus 2; Opus 4.8 78.9% → 74.6% [V56]*
>
> - *LongMemEval chat-memory subset: Opus 4.6 93.1% in Chronos, 76.7% in Claude Code, ~16 points [V23].*
> - *Two administrations of one benchmark, so 15× is a direction, not a measured ratio. Claude Mythos 5 sits above all three at ~88.0% [V56]. [inference]*
>
> *[V23] Is Grep All You Need? · [V56] Harness Engineering.*

Terminal-Bench 2.1 is published two ways, and the pair isolates the variable. [speculative] On the standardised Artificial Analysis run, harness fixed, swapping the model moves three tenths of a point. [V56] Swap only the harness, on the tbench.ai agent-paired board: GPT five-five drops five point two points from Codex CLI to Terminus 2, Opus four-eight four point three from Claude Code to Terminus 2. [V56] Four point three to five point two points from the harness against three tenths from the model is roughly fifteen to one, and the direction is what matters: two administrations rather than one ranking, so read it as an order of magnitude, not a measured ratio. [inference] Which is the decision on this slide — tune the harness before you shop for a model. [speculative] The note's own conclusion: the model is held constant, the harness moves the score. [V56] Across model generations I have no measurement. [inference] And the caveat: the same note puts Claude Mythos 5 above all three, at roughly eighty-eight percent — near the top of the field, which is as far as the note goes. [V56] The sixteen-point figure is a different benchmark — LongMemEval, chat memory, one model inside Chronos against Claude Code,

and Chronos is the authors' own harness. [V23] Scope it: that is a hundred and sixteen of five hundred questions, a twenty-three percent sample whose selection the paper does not discuss. [V23] So measure whether your harness or your model is the variable you have been tuning. [speculative]

12 12. Workflow or agent — almost everything is a workflow

On screen:

> *Part I · Vocabulary*
>
> ***Workflow***
>
> *Orchestrated "through predefined code paths" [S1].*
>
> ***Agent***
>
> *The model "dynamically direct[s] their own processes" [S1].*
>
> - *Across a hand-coded corpus of fifty production loops, the most common configuration is a manually-triggered single agent with a deterministic check, not a swarm [V19]. [inference]*
> - *A workflow does not make failures predictable — your code fixes the tool calls, so you can list every place model output enters the system; an agent picks its path at run time. [inference]*
>
> *[S1] Building Effective Agents — Anthropic. https://www.anthropic.com/engineering/building-effective-agents [V19] Loop Engineering.*

Anthropic's definitions: a workflow runs through predefined code paths you wrote; an agent directs its own process. [S1] In a hand-coded corpus of fifty production loops, the most common configuration is a manually triggered single agent with a deterministic check. [V19] The source says median loop; most common is my wording, because a category has no median. [inference] Why prefer the workflow? Not because it makes failures predictable — it does not. This deck is built by a workflow of fixed steps calling a model at each one, and its content failures were not predictable. [speculative] What it makes is the failure surface enumerable: the tool calls are fixed when you write the code, so you can list every place model output enters the system. [inference] An agent chooses its path at run time, so that list cannot be written in advance. The variance of a single call is identical in both; what changes is what the output can reach. [inference]

13 13. A model grading a model is a checker, not a verifier

On screen:

> *Part I · Vocabulary*
>
> *1*
>
> *Generator*
>
> *Produces the work: the model. [speculative]*
>
> *2*
>
> *Checker*
>
> *Examines the output: a program, a model, a person. Microsoft's alias list leads with evaluator [S14] — there both halves are models, so this deck says checker. [inference]*
>
> *3*
>
> *Verifier*
>
> *Verdict is a function of (output, specification): a compiler, a schema validator. [speculative]*
>
> *4*
>
> *Oracle*
>
> *Tells right from wrong without asking the model [V45]: reference implementation, measurement, person.*
>
> *[S14] AI Agent Orchestration Patterns — Microsoft (2026). https://learn.microsoft.com/en-us/azure/architecture/ai-ml/guide/ai-agent-design-patterns*

A generator produces the work — the model. [speculative] A checker is anything that examines that output: a program, another model, a person. It says nothing about how the verdict is reached, and that weakness is the point. [speculative] A verifier is the strong case: its verdict is a function of the output and a specification. A compiler, a schema validator, a test run. [speculative] An oracle is the other strong case: it tells right from wrong without asking the model. [V45] That definition comes from my predecessor talk's glossary, which requires an oracle to be mechanisable; putting a person on the row is my widening of it, and it is what admits the human expert. [speculative] An oracle you can mechanise becomes a verifier, and verifiers run in CI. [inference] Which places the model judge: a checker and nothing more. Its verdict moves with prompt and sampling, it supplies no ground truth, and it cannot carry a gate. [inference] Now the word itself, because you will not find checker in the documentation. [speculative] The field mostly says evaluator for row two — Anthropic's evaluator-optimizer workflow names it [S1], and Microsoft leads its alias list with the same word [S14]. I am not using it, and the reason is the argument of the whole hour: evaluator covers a compiler and a model's opinion with one word, and the difference between those two is what I am here to draw. [inference] So when you go and read the literature, evaluator is the term to search for; tonight it is checker, so that rows two and three cannot blur. [speculative] Ask which of those three — checker, verifier, oracle — someone has before you accept the word verified. [speculative]

14 14. A shared-transcript multi-agent design bills roughly 15× a chat

On screen:

> *Part I · Cost*
>
> *You pay for every token in and out, on every step of every retry. [inference]*
>
> *The multiplier*
>
> *"Multi-agent systems use approximately 15× more tokens than chats" [V38] — measured once: legacy Opus 4 lead, Sonnet 4 subagents, shared transcript [V58]*
>
> *What moves it*
>
> *Not the model. Caching, compaction and capped-summary sub-agent contracts cut tokens ~38% [V58]*
>
> *15× prices one design, not the category. Ask for the number measured on yours. [V58] [inference]*
>
> *[V38] [V58] Agentic Orchestration Patterns — vault concept note.*

You pay per token in and per token out, on every step, including every retried step. [inference] Now the multiplier everyone quotes: multi-agent systems use approximately fifteen times more tokens than chats. [V38] And now the scope my own note attaches to it, which I want you to carry with the number rather than after it. [speculative] That figure is one benchmark, run with a legacy Opus 4 lead agent and Claude Sonnet 4 subagents, and the architecture is what produces it: a shared transcript, where every subagent carries the whole conversation. [V58] The note's conclusion is that the lever is orchestration design rather than model version — on the same model, caching, compaction and capped-summary contracts between the lead agent and its subagents cut tokens about thirty-eight percent and cost about forty-one. [V58] So treat fifteen times as the price of the un-optimised shared-transcript design, and ask what the design in front of you actually measured. [inference] I raise it here rather than in the cost section because every architectural change I recommend tonight has a token price, and I will keep quoting it. [speculative]

15 15. Reliability is not accuracy: pass^k, not pass@k

On screen:

> *Part I · Vocabulary*

> *Capability and reliability have come apart: solving a task once is not solving it every run. [inference]*
>
> ***pass@k — the ceiling***
>
> *"probability of success in at least one of k attempts" [V44] — what benchmarks report. [inference]*
>
> ***pass^k — the floor***
>
> *Probability that all k attempts succeed — what production actually delivers. [V44]*
>
> - *Rabanser, Kapoor et al. (2026): "recent capability gains have only yielded small improvements in reliability"; "agents that can solve a task often fail to do so consistently" [S6].*
>
> *[V44] Unit Testing — vault note. [S6] Towards a Science of AI Agent Reliability — Rabanser, Kapoor et al. (ICML 2026). https://arxiv.org/abs/2602.16666*

Pass-at-k is the probability of succeeding in at least one of k attempts — that is a ceiling, and it is what demos and benchmark tables report. [V44] Pass-to-the-k is the probability that all k attempts succeed — a floor, and that is what a production system actually delivers. [V44] The two can be very far apart. [inference] A 2026 paper that decomposes agent reliability into twelve separate metrics reports that recent capability gains yielded only small improvements in reliability, and that agents able to solve a task often fail to do so consistently. [S6] Capability and reliability have come apart as axes. [inference] So when a vendor shows you a benchmark score, the follow-up question is not how high — it is how often, and out of how many runs.

16 16. Why a generator

On screen:

> *Part II*
>
> *The negative result, and the four reasons it is narrower than "self-critique never works".*

Part two is the theoretical heart. [speculative] I will make the case as hard as I can for about seven minutes, and then spend three minutes attacking it with the four strongest counterarguments I could find, two of which come from my own notes and one from the researcher whose work the case rests on. [speculative] I do that because a slogan gives you nothing to act on: if you are going to change what you build, or sign off on what somebody else built, you need to know where the claim stops holding. [speculative] The version that survives the attack is narrower, more useful, and still enough to change what you build. [speculative]

17 17. Two loops that look identical from outside

On screen:

> *Part II · Archetypes*
>
> *[code]*
>
> *Same retry code, same logs, same "verified" label — and only the loop that closes on an oracle can come back negative when the generator is wrong. [V3]*
>
> *[V3] Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026).*

Two architectures, and from the outside they are indistinguishable. [speculative] On the left the loop closes on the same model, asked to check its own output. [speculative] On the right the loop closes on an oracle — a compiler, a solver, a schema validator, an external registry — something that can return pass or fail without consulting the thing that produced the artifact. [speculative] The report I built this on states the discriminator: a check tells you something only if it can come back negative when the generator is wrong. [V3] Iteration count does not change that. [inference] Model size does not change that. [inference] When someone shows you an architecture diagram with a box labelled verification, this is the only question worth asking about the arrow that returns.

18 18. Self-rechecking lost accuracy on 3 of 4 tasks

On screen:

> *Part II · Evidence*
>
> *Huang et al.*
>
> *[S3]*
>
> *Self-rechecking lowered GPT-4's GSM8K accuracy*
>
> *ICLR 2024 · two rounds, grade-school maths*
>
> *Stechly et al.*
>
> *[S4]*
>
> *Game of 24 5%→3%, graph colouring 16%→2%; Blocksworld 40% alone, 55% self-checked, 88% external*

> *ICLR 2025 · Blocksworld is block-stacking, program-checkable*
>
> *Vault synthesis*
>
> *[V1]*
>
> *Self-audit 0 of 34; deterministic check 34 of 34*
>
> *2026*
>
> *All three tested a strong generator rechecking with no external signal, not self-critique in general. [inference]*
>
> *[V1] Practitioner Synthesis · [S3] https://arxiv.org/abs/2310.01798 · [S4] https://arxiv.org/abs/2402.08115*

Huang and colleagues, at ICLR 2024: intrinsic self-correction — the model reviewing its own answer with no external signal — lowered GPT-4's grade-school-maths accuracy across two rounds. [S3] Stechly, Valmeekam and Kambhampati, ICLR 2025, on puzzles with machine-checkable ground truth: under self-critique, Game of 24 fell from five percent to three, and graph colouring from sixteen percent to two. [S4] The same paper ran the three-way comparison on Blocksworld — the generator alone at forty percent, the generator checking its own plans at fifty-five, the generator paired with a sound external verifier at eighty-eight. [S4] Self-verification was worth fifteen points there; external verification was worth forty-eight, on the same generator — the cleanest isolation of the verifier as the variable I have. [inference] And the vault synthesis number you already have: zero of thirty-four against thirty-four of thirty-four. [V1] I stress intrinsic: the scope of a negative result matters, and I come back to it in four slides. On multi-agent debate: Huang attributes its gains to self-consistency — samples voting, not self-correction. [S3]

19 19. Why a model cannot check more cheaply than it generates

On screen:

> *Part II · Mechanism*
>
> *Checking is cheaper than producing only when the check is a different procedure. [V12]*
>
> - *Where the intuition holds: solving a sudoku takes trial and error; checking a filled grid is scanning rows, columns and boxes for a repeat. [inference]*
> - *Self-critique's premise: verification "should be easier than generation" [V12].*
> - *Asked to verify, the model runs the same approximate retrieval it ran to generate: that procedure again, not a cheaper one. [V12]*

> • *Scale does not help: better generation brings no "corresponding improvements in its ability to verify its own solutions" [V10].*
>
> *[V10] [V12] Loop Engineering and Self-Verification — vault report.*

The intuition comes from problems where checking is a different job from doing: solving a sudoku takes trial and error, and checking a filled grid means scanning rows, columns and boxes for a repeat. [inference] The report states the assumption directly: the belief that self-critique helps rests on the assumption that verification should be easier than generation. [V12] That holds when checking is a genuinely different computation from producing. [inference] It is not here. [speculative] A model asked to verify runs the same approximate retrieval over the same distribution it just ran to generate: there is no separate checking step, only the same step again. [V12] And the obvious hope does not survive either: improving generation performance does not improve the ability to verify one's own solutions, even on the same task. [V10] Which makes this a design axis rather than a model-procurement one.

20 20. Higher agreement between a generator and its own checker tracks higher mistake vulnerability

On screen:

> *Part II · Mechanism*
>
> *The more a generator and its own checker agree, the more vulnerable the pair is to validated mistakes — my report calls this the consistency dilemma. [V11]*
>
> • *Measured: "higher generator-evaluator self-consistency tracks higher mistake vulnerability — the consistency dilemma" [V11].*
>
> • *And the scaling hope is empirically backwards: "self-verification capability does not improve with generation capability or scale" [V16].*
>
> • *In a system you run: a checker model approves 98 of the generator's 100 outputs. That 98 says the two agree; it says nothing about whether the work is right. [inference]*
>
> *[V11] [V16] Loop Engineering and Self-Verification Loops — vault research report (2026).*

The measured relationship is that higher self-consistency between a generator and its own checker tracks higher vulnerability to validated mistakes — the consistency dilemma. [V11] Read that operationally. [speculative] If you have wired a checker model in front of a generator and the checker approves ninety-eight percent of the time, the natural reaction is that quality is high. [speculative] The measurement says the opposite reading is at least as likely: the two agree because they share the failure structure, and the approval rate is telling you about correlation rather than about correctness. [inference] And the hope that

the next model fixes it is backwards — self-verification capability does not improve with generation capability or scale. [V16] So a high agreement rate between two components of the same system is a diagnostic to investigate rather than a metric to report.

21 21. Self-monitoring fails worst under attack

On screen:

> *Part II · Evidence*
>
> *Indirect prompt injection: an attacker plants instructions in a document or tool result the agent reads; the model executes them. [V57]*
>
> *5×*
>
> *A monitor approving an injected patch: its own output vs. another model's [V9]*
>
> *0.99 → 0.89*
>
> *Rating correct code above incorrect (AUROC), same comparison [V9]*
>
> - *1.0 is perfect, 0.5 a coin toss: one error per hundred pairs becomes one in nine. [inference]*
> - *The failure is adversarial, not random: a monitor sharing the generator's blind spots shares its exploitable ones. [inference]*
>
> *[V9] Loop Engineering · [V57] Harness Engineering — Guardrails.*

One term first, because the slide rests on it. [speculative] In an indirect prompt injection, an attacker plants instructions in what the agent will later read — a document, a web page, a tool's output — and the model executes them on read. [V57] Nothing in the context window separates text to act on from text to read, which is why it works. [inference] Now the measurement. Both numbers compare one model in two conditions: grading output from its own policy, against grading output that is not its own. [inference] Grading its own, it was five times more likely to approve a patch that had followed a prompt injection. [V9] Across the same switch, its ability to rate a correct patch above an incorrect one fell from zero point nine nine to zero point eight nine. [V9] This is not a monitor being a bit noisy. [inference] It is a monitor that shares the generator's blind spots, which means it also shares the ones an attacker can aim at. [inference] If your defence against prompt injection is a model checking a model, you have one component, not two.

22 22. Verification always costs something

On screen:

> *Part II · Limits*
>
> *Every check has a price. [V14]*
>
> - ***A second inference pass*** *— compute and latency each request. [inference]*
> - ***A different model*** *— money and new failure modes. [inference]*
> - ***A deterministic check*** *— engineering time; covers only what you encoded. [inference]*
> - ***A human review step*** *— scarcest; stops scaling first. [inference]*
>
> *I am not claiming nothing can check itself. A 2026 information-theoretic bound on rounds of self-critique is, in my report's words, a diagnostic framework, not an impossibility theorem [V13]. It names what a generator shares with its checker, and forbids no design. [inference]*
>
> *[V13] [V14] Loop Engineering and Self-Verification — vault report.*

Every check has a price — my report puts it as trustworthy knowledge always paying for its own verification — and in a real system you can point at the currency. [V14] There are four. [inference] A second inference pass costs compute and latency on every request; a different model costs money plus a second set of failure modes somebody has to learn. [inference] A deterministic check costs engineering time up front and covers only what you thought to encode; a human review step is the scarcest of the four and the first to stop scaling. [inference] If you cannot point at one of those, you have not bought verification — you have assumed it. [inference] Now the limit, because you may have heard a stronger claim than mine. There is a 2026 information-theoretic bound on rounds of self-critique, and my own report labels it a diagnostic framework rather than an impossibility theorem: the latent failure variable it turns on still needs independent operationalisation before the bound has empirical bite. [V13] So no formal result forbids self-verification. [inference]

23 23. Self-critique degrades when a round adds more errors than it fixes — it is not universally broken

On screen:

> *Part II · Counterarguments*
>
> *Four results qualify "self-critique never works". [inference]*
>
> 1. ***Degradation is a per-model condition*** *— on GSM8K four of seven models degraded, three improved: GPT-5 (96.2%) −1.8 points, Claude Opus 4.6 (97.6%) +0.6 [S10].*
> 2. ***Anthropic says Claude Opus 5 already self-checks*** *— its July 2026 guidance tells developers to remove the "add a final verification step" instruction [V46].*

> 3. ***The verifier can move into training*** *— training-time gains, no inference-time signal [V47].*
> 4. ***LLM-Modulo's authors reject both extremes*** *— Kambhampati et al. reject over-optimism about self-verification and the model-as-mere-translator view alike [S5].*
>
> *[S10] https://arxiv.org/html/2604.22273v2 · [S5] https://arxiv.org/html/2402.01817v2 [V46] Anthropic guidance, vault card · [V47] Digest 2026-07-18*

One: whether it degrades is a per-model condition, and the paper hands you the condition. A 2026 preprint treats self-correction as closed-loop feedback control and derives a steady-state accuracy from two measured rates — how often a round introduces an error, and how often it fixes one — so iterate only while your current accuracy sits below that steady state. [S10] Across seven models on GSM8K, over four rounds, four lost accuracy — GPT-4o-mini minus six point two, GPT-5 minus one point eight, Claude Sonnet 4 minus one point two, GPT-4.1 minus nought point two — and three gained: Claude Opus 4.6 plus nought point six, o3-mini plus three point four, o4-mini flat. [S10] Capability does not predict which: GPT-5 at ninety-six point two percent degrades, Opus 4.6 at ninety-seven point six improves, and the paper's own predictor is the rate at which a round introduces new errors. [S10] One note on the count, because it is easy to check and easy to get wrong: the paper's abstract says GPT-5 and four others lose accuracy, which does not square with its own accuracy table of seven models and three non-degrading. I take the four from the table. [inference] A second preprint covers the other side: on GSM8K-Complex grade-school maths, GPT-3.5, at sixty-six percent baseline accuracy, self-corrected twenty-six point eight percent of its own errors; DeepSeek, at ninety-four, sixteen point seven. [S13] Neither preprint places GPT-3.5 against the other's steady-state condition, so I claim no single curve — only that the effect is set by two rates you can measure, and that a weaker generator has a larger pool of its own errors still left to correct. [inference] Two: Anthropic's July 2026 guidance tells developers to remove the explicit add-a-verification-step instruction for Claude Opus 5, because Opus 5 already re-reads its own output and the extra prompt makes it over-check. [V46] That self-check is a checker by tonight's definitions, not a verifier — which does not blunt the counterargument, only names it. [inference] Scope it: one vendor, one model — Opus 5, not Sonnet 5 or Haiku 4.5 — and no cross-lab replication. [inference] Three: the verifier can move into training — those methods report self-correction gains with no inference-time signal at all. [V47] Four, and this one stings: Kambhampati and colleagues, whose LLM-Modulo framework pairs the model with external verifiers and whose work my case rests on, reject the over-optimistic and the over-pessimistic extreme alike. [S5] I am obliged to sit between them.

24 24. Two claims about self-critique die, two survive

On screen:

> *Part II · Synthesis*
>
> - ***Dies:*** *"Self-critique never works." Three of seven models gained accuracy self-correcting on GSM8K [S10]; GPT-3.5 (66%) corrected 26.8% of its GSM8K-Complex errors against DeepSeek's 16.7% (94%) [S13] — a correction share, not a net gain. [inference]*
> - ***Dies:*** *"Only an inference-time external verifier helps." Training-time verifiers give the same signal, paid once [V47].*
> - ***Survives:*** *A check counts only if it can fail when the generator is wrong — fresh context, another model, outside any model. Say which [V3].*
> - ***Survives:*** *Verification has a price: another pass, another model, a deterministic check, someone's time. Name yours [V14].*
>
> *[S13] Decomposing LLM Self-Correction — arXiv preprint (2026). https://arxiv.org/abs/2601.00828*

Two claims die. [speculative] Self-critique never works fails as a universal: three of the seven models measured gained accuracy from self-correcting — Opus 4.6 plus nought point six, o3-mini plus three point four, o4-mini flat — because their rounds introduced almost no new errors. [S10] On GSM8K-Complex, GPT-3.5 fixed twenty-six point eight percent of its own errors: the share it fixed, not a net accuracy change, and nobody counted the errors self-correction introduced. [S13] Error Depth is the authors' explanation, not a demonstrated mechanism. [S13] And only an inference-time external verifier can help is false too: the verifier can move into training, same signal, paid for once. [V47] Two claims survive. [inference] First: a check tells you something only if it can come back negative when the generator is wrong, so buy that in a fresh context, on another model, or outside any model — and say which one you did. [V3] Second: verification has a price — a second pass, a cheaper model, a deterministic check, someone's time — and if you cannot point at which one you pay, you have not paid it. [V14] That is a weaker thesis than the one on the conference posters. [speculative] It is also the one still true after the next two model releases. [speculative]

25 25. Relabel every self-confirmation step, and measure your checker's agreement rate

On screen:

> *Part II · Decision*
>
> *A model confirming its own output is not a check. [inference]*

> ***Change this week***
>
> - *Find every place your system asks a model to confirm its own output, and label it: not a check. [inference]*
> - *Measure how often your checker model agrees with your generator. Above roughly 90% it is confirming, not checking — seed twenty known defects and count how many it catches. Cannot change the system? Ask for both numbers. [V11] [speculative]*
>
> ***Do not do this***
>
> - *Do not add a second model reading the first one's transcript and call it verified. [V3]*
> - *Do not wait for, or fund, a bigger model. [V10] [V16]*

Do this week: find every point where a system you shipped asks a model to confirm its own output, and relabel it honestly — a formatting step, or a self-consistency step, but not a check. [inference] Second, if you already run a checker model, measure how often it agrees with your generator across a sample of real outputs. Above roughly ninety percent agreement I would read it as confirming rather than checking — it shares the author's blind spots, so its approval carries almost no information, which is the consistency dilemma arriving in your logs. [V11] [speculative] Then test it directly: seed twenty known defects and count how many come back. A checker that agrees constantly and catches few is grading style, not correctness. [speculative] Do not do: adding a second model that reads the first one's transcript and then describing the system as verified — the transcript is the contamination. [V3] And do not wait for the next model, because verification capability does not track generation capability. [V10] [V16] Now, what actually works.

26 26. What an external check buys

On screen:

> *Part III*
>
> *The positive half, with the bill and the blind spots attached.*

Part three is the constructive half, and it has the strongest number in the talk in it. [speculative] I will show what closing a loop against an external check delivers, then the same result reproduced in three unrelated fields, then the three things it does not do and the two ways it costs money. [speculative] I sequence it that way deliberately: a technique presented without its boundary conditions is a sales pitch, and this room has sat through enough of those. [speculative] The boundary conditions are also where the interesting engineering is. [speculative]

27 27. An external verifier moved structural code compliance from 56.8% to 98.6%

On screen:

> *Part III · Anchor*
>
> *56.8% → 98.6%*
>
> *Open loop, then closed loop [V4] [S7]*
>
> - *The authors credit "closing the generation–verification loop with an external verifier" [V5].*
> - *Model-invariant: swapping the backbone model, DeepSeek to Claude, did not move compliance [V5]. Safety-critical structural design [V4] is my own field, so treat me as an interested party. [speculative]*
> - *Blocksworld, a planning benchmark: 40% alone → 55% self-checked → 88% with a sound external verifier — self-check +15 points, external verifier +48, same generator. [S4]*
>
> *[V4] [V5] Loop Engineering and Self-Verification — vault report · [S7] https://arxiv.org/abs/2608.07978 [S4] Self-Verification Limitations — Stechly et al. https://arxiv.org/abs/2402.08115*

This is the number that made me rewrite my own architecture advice. [speculative] A closed-loop multi-agent framework moved structural code compliance from fifty-six point eight to ninety-eight point six percent. [V4] Same agents. [V5] The authors state the attribution themselves: the gain does not come from a more capable model, it comes from closing the generation-verification loop with an external verifier. [V5] And the result is model-invariant — swap the backbone model from DeepSeek to Claude and compliance does not move detectably, because the guarantee lives in the verifier rather than in the generator. [V5] The domain is safety-critical structural design, which is my own field, so treat me as an interested party on this number. [speculative] What transfers is the model-invariance: it is the signature of a guarantee that lives outside the model. [inference] What does not transfer is the field, which is why the second number is here. [speculative] Blocksworld is a planning benchmark with machine-checkable ground truth — a synthetic research task rather than any industry, and not engineering: one generator scored forty percent alone, fifty-five checking its own plans, eighty-eight paired with a sound external verifier. [S4] Same ordering, different discipline. [inference]

28 28. One architecture in three fields: the generator proposes, an oracle outside it returns pass or fail

On screen:

> *Part III · Convergence*
>
> *Structural design*
>
> *[V4]*
>
> *56.8% → 98.6% structural code compliance, external physics verifier*
>
> *Model-invariant*
>
> *Clinical diagnosis*
>
> *[V7]*
>
> *"clinical-diagnosis accuracy from 55.6% to 77.5%"*
>
> *Guideline oracle*
>
> *Formal mathematics*
>
> *[V8]*
>
> *422% improvement over the best publicly available prover LLM*
>
> *Proof-checker oracle; vendor blog, metric unnamed*
>
> *All three used an oracle the field already owned. Name yours. [inference]*
>
> *[V4] [V5] [V7] [V8] Loop Engineering and Self-Verification Loops — vault report. Only the structural study reports a backbone-model swap [V5].*

Structural design: fifty-six point eight to ninety-eight point six, with a physics verifier. [V4] Clinical diagnosis: fifty-five point six to seventy-seven point five, grounded against expert clinical guidelines. [V7] Formal mathematics: a four-hundred-and-twenty-two percent improvement over the best publicly available prover model, using a proof checker as the oracle. [V8] Different domains, different oracle types, and one architecture underneath: a generator proposes, something outside it returns pass or fail without reading how the answer was produced. [inference] Only the structural study swapped its backbone model, and the gain survived; the other two report no such comparison, so model-invariance belongs to that one study. [V5] What all three share is an oracle the field already had: a physics solver, a clinical guideline, a proof checker. [inference] So look in your own field for what already returns pass or fail. [speculative]

29 29. Independence costs a fresh context, not a second vendor

On screen:

> *Part III · Independence*
>
> *Run the check on the same model in a clean context: the artifact and criteria only. [V17]*
>
> - *"checker independence is a context-isolation boundary, not a model-weights boundary" [V17] — buy a second model family only for the subjective checks no program can make.*
> - *The transcript contaminates it: a checker that sees the generator's reasoning is biased toward confirming it. [V17]*
> - *Rule: pass the artifact and the specification, never the chain of thought. [inference]*
> - *Ceiling: same-family checkers share blind spots — "independence is approximated, never absolute" [V54].*
>
> *[V17] [V54] Loop Engineering — vault concept note (2026).*

Independence is not what most teams assume. [speculative] Checker independence is primarily a context-isolation boundary rather than a model-weights boundary. [V17] You do not need a second vendor for the independence that matters most of the time — the note reserves a second model family for the subjective checks no program can make, and nothing else. [V17] The same model, in a fresh context window, shown only the artifact and the acceptance criteria, is where the load is carried. [V17] What destroys it is the transcript. [inference] A checker that can read the generator's reasoning is biased toward confirming that reasoning. [V17] So the implementation rule is one line: pass the artifact and the specification, never the chain of thought that produced it. [inference] That costs one extra inference call and some discipline. [inference] Now the ceiling: the same note records that same-family checkers retain shared blind spots and can collude in a degeneration-of-thought mode, so independence is approximated, never absolute. [V54] Context isolation buys most of what is available; no configuration buys all of it. [inference]

30 30. Five verification levels — name the one that checked your last AI-assisted decision

On screen:

> *Part III · Levels*
>
> *1*
>
> *Deterministic*
>
> *Exit code, assertion, known-good result. [V18]*
>
> *2*

> *Rule or constraint*
>
> *Linter, schema, policy engine. [V18]*
>
> *3*
>
> *Field truth*
>
> *Tests, a deploy, a real outcome. [V18]*
>
> *4*
>
> *Model as judge*
>
> *Rubric scoring by a model. [V18]*
>
> *5*
>
> *Human checkpoint*
>
> *A person with a stake. [V18]*
>
> *Most production checking is a program: "70% of loops verify at levels 1–2" [V18].*

Five levels. [speculative] One, deterministic: an exit code, an assertion, a known-good result. [V18] Two, rule or constraint: a linter, a schema, a policy engine. [V18] Three, delayed field truth: tests, a deploy, a real customer outcome you observe later. [V18] Four, model as judge: a model scoring against a rubric. [V18] Five, a human checkpoint where someone with a stake signs. [V18] Across the measured corpus, seventy percent of loops verify at levels one and two. [V18] I want that to land as good news rather than as immaturity. [speculative] The boring levels are cheap, fast, auditable, and they are where measured production checking actually happens. [V18] The question to carry out of this room is exactly four words: which level verifies this?

31 31. Verification level 4 is coverage, not certification

On screen:

> *Part III · Levels*
>
> *Level 4 of the five-level verification scale is a model scoring against a rubric [V18]. [inference]*
>
> ***What level 4 gives you***
>
> *Coverage of what no program can express — tone, relevance, whether an answer addressed the question. [speculative]*
>
> ***What level 4 cannot buy***

> *Independence from failures it shares with the generator. The consistency dilemma: the more a generator and its own checker agree, the more vulnerable the pair is to mistakes both validate. [V11]*
>
> - *Ordering: make the check objective, isolate the checker's context, then change model family. [V17] [V18]*
> - *If a decision only reaches level 4, say so in the design review. [inference]*

Level four buys coverage of things no program can express: tone, relevance, whether the answer addressed the question. [speculative] What level four cannot buy is independence from the failures it shares with the generator — the consistency dilemma from part two: the more a generator and its own checker agree, the more vulnerable the pair is to mistakes both of them validate. [V11] So the ordering: make the check objective if you can, isolate the checker's context if you cannot, and only then change model family — each step costs more and buys less. [V17] [V18] [inference] And if a decision only reaches level four, say so in the design review rather than writing verified on the diagram.

32 32. Grounding is not the same thing as retrieval

On screen:

> *Part III · Grounding*
>
> *1*
>
> *Parametric*
>
> *From training alone. No evidence, no constraint. [V26]*
>
> *2*
>
> *Structural*
>
> *Retrieval narrows the output distribution; it certifies nothing. [V26]*
>
> *3*
>
> *Deterministic*
>
> *A program outside the model computes the answer — a constraint solver, a maths library, a ledger reconciliation. The model only formalises the input. [V26]*
>
> - *Layer three nears the "mere translator" extreme LLM-Modulo's authors decline. [S5] [inference]*
>
> *Most products marketed as grounded stop at layer two. [speculative]*

> *[V26] Structural Grounding — vault concept note.*

Layer one: the model answers from its training weights, with no evidence and no constraint. [V26] Layer two: retrieval — you inject documents, and the output distribution narrows toward them. [V26] That is a genuine improvement, and my read — nobody has published a count — is that most products marketed as grounded stop there. [speculative] Layer three: a computation outside the model determines the answer, and the model extracts, selects and translates but never performs the truth-determining computation itself. [V26] Now the attack on that third sentence, which is stronger than the field's most careful researchers will go. [speculative] The LLM-Modulo authors — the group behind the Blocksworld and self-verification results in part two — name two errors, and the second is treating the model as a mere translator of the problem specification. [S5] Layer three as worded sits close to that extreme, so take it as a design target rather than a settled result. [inference] Hold the layer distinction: the gap between two and three is where the next slide lives.

33 33. A faithfulness score cannot catch a wrong source

On screen:

> *Part III · Grounding*
>
> *A system can quote its source perfectly and still be structurally unsafe. [V15]*
>
> - *Worked case: an assistant retrieves a clause and reproduces it exactly — and the clause was wrong. [V15]*
> - *The scoring: "it scores 90%+ on faithfulness (it matched its retrieved source exactly) and is structurally unsafe, because no output-against-context check can catch a wrong context" [V15].*
> - *Faithfulness measures agreement with the retrieved document, never with reality. [inference]*
> - *The question that catches it: not "did it match the source" but "who decided that was the source?" [inference]*
>
> *[V15] Loop Engineering — Practitioner Synthesis — vault report.*

Picture an assistant that retrieves a regulatory clause and reproduces it word for word — and the clause is wrong. [V15] On every faithfulness metric in common use it scores above ninety percent, because faithfulness measures whether the output matched the retrieved source, and it did, exactly. [V15] The system is also unsafe, and no output-against-context check can catch it, because the check compares the answer to the context and the context is the thing that is wrong. [V15] That gap between faithful and true

is where the most expensive failures live, and they are invisible to the metric most vendors will show you. [inference] So the audit question is not did it match the source. [inference] It is who decided that was the source, and what verified that decision. [inference]

34 34. Boundary one — the oracle only sees what it measures

On screen:

> *Part III · Limits*
>
> *An external check supplies information about its own dimensions and nothing else. [V6]*
>
> - *The structural-design study behind this talk's headline number — structural code compliance 56.8% → 98.6% with an external verifier [V4] — states its own limit: "when the defect originates from the structural topology, the parameter-level closed loop is power-less" [V6].*
> - *Generalised: a schema validator cannot catch a wrong schema; a test suite cannot catch a missing requirement. [inference]*
> - *So the design question is coverage, not presence — "which failures does this oracle not see?" [inference]*
>
> *[V4] [V6] Loop Engineering and Self-Verification Loops — vault research report (2026).*

The structural-design study behind my headline number — fifty-six point eight to ninety-eight point six percent, with an external verifier [V4] — states its own limit: when the defect originates from the structural topology, the parameter-level closed loop is powerless. [V6] In plain terms, the verifier checked whether the parameters were compliant, so it could say nothing about whether the overall shape was wrong. [inference] A schema validator cannot catch a wrong schema. [inference] A test suite cannot catch a requirement nobody wrote a test for. [inference] A citation registry can confirm a paper exists and cannot confirm it says what was claimed. [inference] So the design question is never does this have a verifier — it is which failures does this verifier structurally not see, and what covers those. [inference]

35 35. Boundary two — somebody has to write the verifier

On screen:

> *Part III · Limits*

> *The verifier may be more work than the generator, and no source I have puts a figure on it. [inference]*
>
> - *LLM-Modulo is Kambhampati et al.'s architecture: the model generates candidates, external verifiers and solvers check them. Their caveat — that verifier may be substantial work, and substituting a simulator means writing the simulator. [S5]*
> - *The work is formalising domain knowledge into machine-checkable form — "labor-intensive and domain-specific" [V26] — which a vendor cannot cheaply copy. [speculative]*
> - *Hardest where it is most needed: "no convincing public playbook yet for retrofitting a ten-year-old codebase" [V22].*
>
> *[S5] https://arxiv.org/html/2402.01817v2 · [V22] Harness Engineering · [V26] Structural Grounding*

I have been filing verifier construction under open problems. [speculative] It is not an open problem — it is the bill. [inference] Kambhampati and colleagues propose LLM-Modulo — the model generates candidates, external verifiers and solvers check them — and they put its bill bluntly: the verifier may be more complex and involve substantial work, and substituting a simulator for a verifier means someone writes that simulator. [S5] The work is formalising domain knowledge into machine-checkable form, which my own notes call labor-intensive and domain-specific. [V26] Expensive, and defensible: whoever formalises a domain owns the check for it, and a generic vendor cannot cheaply copy that. [speculative] And the hardest case arrives with an existing codebase: there is no convincing public playbook yet for retrofitting a ten-year-old codebase. [V22]

36 36. Boundary three — the verifier depreciates

On screen:

> *Part III · Limits*
>
> *Treat a verifier as "a maintenance liability that must be re-scoped at every model upgrade" [V25].*
>
> - *Reported first-hand: a checker "became unnecessary overhead" for tasks that moved inside the generator's boundary after a model upgrade [S2].*
> - *The same account deleted a whole context-reset subsystem after one upgrade [S2].*
> - *Plan for it: put a re-scope of every model-judge step on the calendar for each model upgrade. [inference]*
> - *A compiler or a schema validator does not need re-scoping when a model ships; a model-judge step does. No source puts a rate on either. [inference]*

> *[V25] Verification and Checker Design — vault report · [S2] https://www.anthropic.com/engineering/harness-design-long-running-apps*

Everything I am telling you to build depreciates on a model-release cadence. [inference] My own notes put it precisely: a verifier is not a fixed asset but a maintenance liability that must be re-scoped at every model upgrade. [V25] There is a first-hand engineering account of exactly this — a checker became unnecessary overhead once the generator absorbed the check after a model version bump, and the same team deleted an entire context-reset subsystem after a later model version made it redundant. [S2] So put the re-scope on the calendar rather than discovering it in the token bill. [inference] And notice which levels survive: a compiler does not depreciate when a model ships. [inference] A model-judge step does. [inference] That is a third reason to spend your budget at verification levels one and two — an exit code, a schema — rather than on a model judge.

37 37. What to do when you have no oracle

On screen:

> *Part III · Limits*
>
> *Most systems already contain a check nobody wrote down [speculative]; where none exists, the honest engineering answer is to leave that decision un-automated. [S1]*
>
> ***You have one, probably***
>
> - *A schema, a registry, a reconciliation, a delayed real-world outcome, a person with a stake. [V18]*
> - *Report the reliability floor rather than the ceiling: pass^k, not pass@k. [V44]*
>
> ***You genuinely do not***
>
> - *Then the honest answer is to not automate that decision. [S1]*
> - *Have it answer "cannot verify" and route to a person; test that path. [speculative]*
>
> *[S1] Building Effective Agents — Anthropic. https://www.anthropic.com/engineering/building-effective-agents [V18] Loop Engineering · [V44] Unit Testing.*

The obvious objection to everything so far: this whole framework assumes you have an oracle, and plenty of decisions have no compiler. [speculative] First, most people underestimate what they have — a schema, a reconciliation against a second system, a registry lookup, a delayed real-world outcome you could actually observe, or a person with a stake who signs. [V18] Those are levels one through five, and at minimum you can report the reliability floor honestly: all-k-succeed rather than at-least-one-succeeded. [V44] Then the honest engineering answer is to not automate that decision, and the vendor guidance most

quoted in this field says the same thing — find the simplest solution, which might mean not building an agentic system at all. [S1] That is an unsatisfying slide for a room that came for what to build. [speculative] It is also the one that saves the most money. [speculative]

38 38. Write down your oracle, isolate the checker's context, list the blind spots

On screen:

> *Part III · Decision*
>
> *Three writing exercises, none of which needs a new tool or a new model. [speculative]*
>
> *1*
>
> *Name the oracle*
>
> *For each AI-assisted decision, write down what could say pass or fail without the model. [V18]*
>
> *2*
>
> *Isolate the context*
>
> *Checker sees artifact plus criteria; never the generator's transcript. [V17]*
>
> *3*
>
> *Write the blind spots down*
>
> *Beside each oracle, the failures it structurally cannot see. [V6]*
>
> *[V6] [V17] [V18] Loop Engineering — vault report and concept note (2026).*

One: for each AI-assisted decision in your product, write down what could return pass or fail without consulting the model — that is your oracle, and an empty box means that decision has nothing that can come back negative when the model is wrong: it ships on the model's output alone. Build the check there first, because that is the decision where a wrong answer reaches production unchallenged. [V18] Two: wherever you already run a checking step, isolate its context so it sees the artifact and the acceptance criteria and never the transcript that produced them; that change is usually an afternoon. [speculative] Three, and this is the one teams skip: next to each oracle write the failures it structurally cannot see, because that list is your actual risk register and it is far more informative than a coverage percentage. [V6] Now, authority — which is where the expensive incidents live.

39 39. Bounded autonomy

On screen:

> *Part IV*
>
> *The nine-second database deletion, the permission decisions behind it, and the five autonomy levels.*

Part four, and I want to reframe the topic before I start. [speculative] Autonomy gets discussed as an AI safety question, in language borrowed from alignment research, and that framing makes the problem look novel and slightly unreachable. [speculative] I think it is mostly a permissions question, and permissions engineering has fifty years of results behind it — so I will name the governing principle from 1975, because this room spans people who apply it weekly and people who have never had to. [speculative] So the argument in this section is that the interesting failures are not the model doing something surprising — they are the model being able to do something nobody had bounded. [speculative] I will start with the arithmetic that explains why long chains fail, then the incident that explains why authority matters more than the arithmetic. [speculative]

40 40. Why 95% per step still fails over twenty steps

On screen:

> *Part IV · Arithmetic*
>
> *End-to-end success is the product of the per-step probabilities: the chain fails on length. [V27] [inference]*
>
> *$0.95^{20} \approx 36\%$*
>
> *20 steps at 95% each [V27]*
>
> - *Only shortening the chain touches the exponent; checking each seam catches the failed step there, not downstream. A better model only nudges p_i. [inference]*
> - *The product assumes per-step accuracies are independent; real ones correlate. Of its own ten-step version, $0.95^{10} \approx 0.60$, my research brief says "the number is a slogan, not a prediction" [V28] — 20 steps inherits that. [inference]*
>
> *[V27] Agentic Loop Pattern · [V28] Grounding in AI-Driven Systems.*

Ninety-five percent per step, twenty steps, and you are at thirty-six percent end to end. [V27] Ninety-five percent sounds excellent in a status meeting and is catastrophic in a chain. [inference] Shortening the chain is the only move on the exponent — it removes factors from the product. [inference] Checking each seam is different: a failed step is caught and retried there rather than multiplied downstream. [inference] A better model does neither; it nudges each p-sub-i, and twenty still multiply. [inference] Now the caveat, because the number gets quoted without it constantly. [speculative] The product assumes the per-step accuracies are independent, and in real pipelines they are not: a hard query gets bad retrieval and bad generation at once, so the errors correlate. [V28] My own brief is blunt about that — the number is a slogan, not a prediction — and to be exact about what I am borrowing, it writes that about its own ten-step version, ninety-five percent per step giving about sixty percent end to end. [V28] It never discusses the twenty-step case. The carry-over is mine: same series product, same independence assumption, so the same defect. [inference] Use either number to reason about shape, never to forecast a specific system.

41 41. Real pipeline errors are not independent — they stick

On screen:

> *Part IV · Arithmetic*
>
> *Errors in real pipelines are not independent — they stick and propagate. [V29]*
>
> - *Work on compounding errors in agentic pipelines reports failures "propagating semantically across steps rather than occurring independently" [V29].*
> - *A separate study, DSAP, measured recovery per stage on 99 SWE-Bench instance-arm pairs (bug fixing): "37.5% for test generation but 0% for patch generation" [V29].*
> - *So the curve is not a clean exponential: stages with different recovery, some with none at all. [inference]*
> - *Practical consequence: instrument per stage. An end-to-end success rate hides which seam is leaking. [inference]*
>
> *[V29] The 4/δ Bound — Designing Predictable LLM-Verifier Systems — vault reading note.*

Work on compounding errors in agentic pipelines reports failure stickiness — errors propagating semantically across steps rather than independently — so the independence assumption fails in the direction that hurts. [V29] A separate study, DSAP, measured recovery per stage on SWE-Bench, a bug-fixing benchmark: thirty-seven and a half percent for test generation, zero for patch generation. [V29] Zero. [speculative] So the true picture is not a smooth exponential decay; it is a set of stages with wildly different recovery behaviour, some of which never recover once they go wrong. [inference] Measure per stage, not end to end, because an end-to-end success rate averages over exactly the difference that would tell you where to put your one check. [inference]

42 42. Nine seconds: an agent deleted a production database

On screen:

> *Part IV · Incident*
>
> *Date*
>
> *25 April 2026 [V30]*
>
> *What happened*
>
> *A coding agent "deleted PocketOS's entire production database and all volume-level backups in nine seconds" [V30]*
>
> *How*
>
> *A credential mismatch on a staging task; a token found in an unrelated file; a destructive API call with no confirmation step [V30]*
>
> *Cost*
>
> *A 30-hour outage for the businesses running on it [V30]*
>
> *[V30] PocketOS Cursor Incident — Database Deletion in Nine Seconds — vault watch note (2026).*

On the twenty-fifth of April this year, a coding agent deleted a company's entire production database and all volume-level backups in nine seconds. [V30] The sequence: it hit a credential mismatch on a staging task, found an API token sitting in an unrelated file, and issued a destructive call against an API that had no confirmation step for destructive operations, no environment isolation, and backups stored on the same volume as the primary data. [V30] Read that list again and notice how much of it has nothing to do with artificial intelligence. [speculative] A token with too much scope, no environment separation, backups on the same volume, no confirmation on a destructive operation. [inference] That is a 1975 finding with a 2026 date on it. [speculative]

43 43. Ask why the agent was able to delete the database

On screen:

> *Part IV · Incident*

> *Nothing in the PocketOS deletion required the model to misbehave: an over-scoped token and an unconfirmed destructive API are enough. [V31] [inference]*
>
> - *The post-mortem's conclusion: "system prompts are not security controls" [V31] — prevention needs a deterministic mechanism outside the reasoning loop.*
> - *The 1975 principle: every program and user runs with the least privilege needed for the job [S9].*
> - *A token in an unrelated file carried authority over production volumes — a scoping defect, not a model defect. [inference]*
>
> *[V31] PocketOS Cursor Incident — vault watch note · [S9] The Protection of Information in Computer Systems — Saltzer and Schroeder (1975). https://www.cs.virginia.edu/~evans/cs551/saltzer*

Everyone asks why did the agent do that. [speculative] The useful question is why was it able to. [inference] The most rigorous post-mortem of that incident states the conclusion I would put on a wall: system prompts are not security controls. [V31] The only thing that reliably prevents a catastrophically wrong action is a deterministic enforcement mechanism operating outside the model's reasoning loop, one that makes certain outcomes structurally impossible. [V31] Saltzer and Schroeder wrote the governing principle in 1975 — every program and every user operates with the least set of privileges necessary for the job. [S9] A token found in an unrelated file carried authority to delete production volumes. [inference] No prompt engineering fixes that, and no better model fixes that. [inference] A narrower credential fixes it. [inference]

44 44. Deny at the API gate, not in the prompt

On screen:

> *Part IV · Design*
>
> *A gateway rule is evaluated after the model and refuses the emitted call whatever the model produced; a prompt is input that the next tool result can override. [V32] [inference]*
>
> ***Advisory — tool output can override it** [inference]*
>
> *Prompt, system message, policy document. [inference]*
>
> ***Enforced — holds whatever the model emitted** [inference]*
>
> *The credential, the API gateway, the sandbox, the deny gate. [V32]*
>
> - *Verbatim: "irreversible actions (production deploys, money movement, force-push) belong on a deterministic deny gate, never on a probabilistic reviewer" [V20].*

> *[V32] Harness Engineering — Tool Permissions — vault report (2026). [V20] Loop Engineering — vault note (2026).*

My source states the design rule: put the control where the authority is exercised, not where the intent is expressed. [V32] A prompt instruction is advisory — anything that changes the input can override it, and for an agent reading tool output that is essentially every turn. [inference] A credential, an API gateway rule, a sandbox or a deny gate is evaluated after the model and outside it, so it holds whatever the model produced. [inference] The operational form, verbatim: irreversible actions — production deploys, money movement, force-push — belong on a deterministic deny gate, never on a probabilistic reviewer. [V20] This also separates two components people conflate: [speculative] the verifier checks the work, the gate checks the action, and you need both. [inference]

45 45. Anthropic's OS sandbox cut Claude Code permission prompts by 84%

On screen:

> *Part IV · Design*
>
> *Anthropic hardened Claude Code's boundary at the OS and reports 84% fewer permission prompts — approval reduced, never removed. [V55]*
>
> *Why*
>
> *"containment over supervision", because "human oversight suffers from approval fatigue" [V55]*
>
> *What*
>
> *OS-level sandbox — Seatbelt, bubblewrap: "writes inside workspace, network denied by default" [V55]*
>
> *What still escapes*
>
> *Same post: models "helpfully" escaped sandboxes or routed around restrictions [V55]*
>
> *[V55] How We Contain Claude — Anthropic (2026), vendor-reported, via vault watch note.*

Anthropic shifted much of Claude Code's control off per-action human approval and onto the operating system — Seatbelt on macOS, bubblewrap on Linux, reads allowed, writes confined to the workspace, network denied by default. [V55] Their stated reason is exactly this section's argument: containment over supervision, because human oversight suffers from approval fatigue. [V55] They report eighty-four percent fewer permission prompts in Claude Code — fewer, not none, and the post never says what share of actions still needs a human yes. [V55] And the same post discloses the leak — documented cases of

models escaping sandboxes or routing around restrictions. [V55] So a sandbox is a boundary you maintain rather than one you install. [inference] That is a vendor describing its own product, cited from a vault summary rather than the post itself, and still the most concrete containment reference I have. [V55]

46 46. A guardrail checks one channel — NeMo's content rails never read a tool call's arguments

On screen:

> *Part IV · Design*
>
> *A guardrail inspects a named subset of traffic; the rest is unchecked. [inference]*
>
> - *NVIDIA's NeMo Guardrails checks a tool call is declared and schema-valid, but its content rails read message content only: argument payloads go uninspected [V33] [S15].*
> - *And "tool results bypass input rails" — tool output is trusted by default [V33].*
> - *Stack layers that fail differently: input rail, argument check, deterministic deny gate on irreversible actions. [V33] [V20]*
>
> *Ask your guardrail vendor what its filter skips; that is your unprotected surface. [inference]*
>
> *[V33] Guardrails and Runtime Enforcement — vault report. [S15] NeMo Guardrails: Tool Calling — NVIDIA. https://docs.nvidia.com/nemo/guardrails/configure-guardrails/guardrail-catalog/tool-calling*

Guardrail frameworks are useful and I run them, but read the documentation: it says where they stop. [speculative] A rail is one check on one channel: the incoming message, the model's output, or a tool call. [inference] NVIDIA's guide for NeMo Guardrails discloses that its content rails evaluate the message content field only, so an argument payload is never inspected, even when the call itself passes the schema check. [V33] [S15] And tool results bypass input rails, so tool output is trusted by default and can steer everything after it. [V33] Those are not bugs but documented scope — NVIDIA wrote both gaps down. [inference] So stack layers whose holes do not line up, and never let a diagram describe a single rail as protection. [inference]

47 47. Five autonomy levels, and each names who answers

On screen:

> *Part IV · Autonomy*
>
> *1*
>
> *Human in the loop*
>
> *Every action approved. [V34]*
>
> *2*
>
> *Human on the loop*
>
> *Acts inside pre-authorised bounds; monitor interrupts. [V34]*
>
> *3*
>
> *Supervised*
>
> *Routine handled; high-stakes escalate. [V34]*
>
> *4*
>
> *Delegated*
>
> *End-to-end in mission bounds; audited after. [V34]*
>
> *5*
>
> *Full autonomy*
>
> *Own objectives; governance and audit. [V34]*
>
> *"Each level implies a distinct accountability structure" [V34] — the level picks who answers. [inference]*

One through five run from every action approved, to a system operating on objectives set inside broad constraints with the human role reduced to governance and audit. [V34] One caveat on those names. [speculative] They are the vault note's own synthesis, and the note says so. [V34] The paper underneath names its five levels operator, collaborator, consultant, approver and observer, and centres them on the role the user takes rather than on the system's scope, so the mapping between the two is approximate. [V34] I use the synthesis because it names the accountability holder directly, which is the decision you are actually making. [speculative] The load-bearing sentence is that each level implies a distinct accountability structure. [V34] So this is not a maturity ladder to climb. [inference] Choosing a level is choosing who is answerable when it goes wrong — a decision written down, not an emergent property of a tool default.

48 48. Human approval is a control that degrades

On screen:

> *Part IV · Autonomy*
>
> *Bainbridge*
>
> *[S8]*
>
> *The operator monitors a system installed because it outperforms them*
>
> *1983*
>
> *The Code*
>
> *[V35]*
>
> *Developers overseeing AI agents: "those skills atrophy" — 17% drop in skill mastery*
>
> *Newsletter*
>
> *RSNA*
>
> *[S12]*
>
> *Radiologists, 15+ years: 82% → 45.5% on a purported AI's incorrect suggestion*
>
> *RSNA meeting summary, 2023*
>
> *Measure your approver's override rate: zero changed or rejected is a signature, not a control. [inference]*
>
> *[S8] https://ckrybus.com/static/papers/Bainbridge_1983_Automatica.pdf · [S12] https://www.rsna.org/news/2023/may/ai-bias-may-impair-accuracy [V35] Agentic Coding Is a Trap — The Code.*

Whenever anyone says keep a human in the loop, they are proposing a control whose reliability is almost never measured. [inference] Lisanne Bainbridge described the problem in 1983, in the Ironies of Automation: the automation was installed because it outperforms the operator, and then the operator is asked to monitor it. [S8] That is the task humans are worst at. [inference] A 2026 newsletter reports that the skills required to supervise decay from not exercising them — a seventeen percent drop in skill mastery, with debugging declining most steeply. [V35] And in a domain of certified experts, radiologists with more than fifteen years of experience went from eighty-two percent accuracy to forty-five point five when the purported AI suggested the incorrect category. [S12] Purported is the report's own word: the suggestions were experimenter-controlled and only labelled as AI, which makes the thirty-six-and-a-half-

point drop an effect of the label rather than of any model. [inference] Bainbridge in 1983, the radiology result in 2023 — forty years apart, same finding. [inference] A human approval step is a real control, and it is not free.

49 49. The EU AI Act says what a human approver must be able to do

On screen:

> *Part IV · Autonomy*
>
> *Article 14 asks more of an approver than a signature. [S17]*
>
> - *Article 14(4), high-risk systems: the overseer must be enabled, "as appropriate and proportionate", to stay aware of over-relying on the output, and to override and halt the system [S17].*
> - *One reading of Annex III, the high-risk list — a vendor guide, not a regulator: classification turns on whether the system "materially influences decisions", not on a counter-signature [S16].*
> - *Measure the override rate: decisions changed, rejected or halted. Zero is counter-signing, not oversight. [inference]*
>
> *[S17] EU AI Act Article 14. https://artificialintelligenceact.eu/article/14/ [S16] Annex III HR guide — Knowlee. https://www.knowlee.ai/blog/ai-act-annex-iii-hr-employment*

Article 14 is the human-oversight article, and it applies to high-risk systems — so this is a compliance floor for that class, not a general rule. [S17] Paragraph 4b is worth reading out loud: whoever is assigned oversight must be enabled, as appropriate and proportionate, to remain aware of the tendency to over-rely on the output — automation bias, in the legal text. [S17] Paragraphs 4d and 4e add that the same person must be able to disregard, override or reverse the output, and interrupt the system. [S17] So the radiology number is the failure mode the article legislates against. [inference] And on one published reading — a vendor guide rather than a regulator, covering employment systems only — a counter-signature does not by itself take a system out of Annex III classification, because classification turns on whether the system materially influences the decision. [S16] Take the reasoning, not the coverage. [S16] So if your verification level is a person clicking approve, measure the override rate: decisions that approver changed, rejected or halted. [inference] Zero overrides across a review period is counter-signing, not oversight. [inference] Test it: seed known-bad cases into the queue and count how many the approver stops. [inference]

50 50. Bounded autonomy can relocate blame instead of risk

On screen:

> *Part IV · Counterargument*
>
> *Putting a named person at the boundary can move the liability without moving the risk. I have no clean answer. [speculative]*
>
> - *The role has a name — the moral crumple zone: a person absorbing blame for "a system they can't actually override" [V37].*
> - *We cite the model's error rate as proof it cannot certify itself, then install a human whose error rate nobody measures. [inference]*
> - *Partial answer: set the autonomy level so the human's task stays doable, and measure that. [inference]*
> - *Honest answer: if the person cannot realistically override the system, the level is wrong. [speculative]*
>
> *[V37] 2026-07-22 Digest — vault journal note.*

When you place a named human at the boundary, you may not be reducing risk — you may be relocating liability. [inference] The term for the role is the moral crumple zone: the nearest human absorbs blame for a system they cannot actually override, while the technology's reputation stays intact. [V37] Notice the inconsistency in my own argument. [speculative] I spent twenty minutes citing measured model error rates as proof that a model cannot certify itself, and then I proposed a human oracle whose error rate nobody in this room measures. [inference] Partial answer: choose the autonomy level so the human's task remains something a human can actually perform, and then measure whether they perform it. [inference] Honest answer: if the person cannot realistically override the system, you have set the level wrong and the sign-off is decoration. [speculative]

51 51. Gate the irreversible actions, scope the credentials, name who answers

On screen:

> *Part IV · Decision*
>
> *Three permission changes to make, and three habits that only look like controls. [speculative]*

> ***Do this***
>
> • *Enumerate irreversible actions; put each behind a deterministic gate. [V20]*
>
> • *Scope every agent credential to the one task it serves. [S9]*
>
> • *Write down the autonomy level per feature, with the name of who answers. [V34]*
>
> ***Stop doing this***
>
> • *Relying on a system prompt as a control. [V31]*
>
> • *Describing a single guardrail as protection. [V33]*
>
> • *Counting an approval click as oversight without measuring it. [S17]*
>
> *Cannot change the system? Ask for one number: the share of decisions the approver changed, rejected or halted. [S17] [inference]*

Start: enumerate every irreversible action your system can reach and put each one behind a deterministic gate, and that list is shorter than most teams expect. [V20] Enumerating it is an afternoon. [speculative] Scope every agent credential to exactly one task, which is the direct fix for the incident I described. [S9] And write down the autonomy level for each feature together with the name of the person who answers for it. [V34] Stop: relying on a system prompt as a control, because it is not one. [V31] Stop describing a single guardrail as protection when its own documentation lists what it does not inspect. [V33] And stop counting an approval click as oversight unless you are measuring whether anyone ever declines. [S17] Now the money.

52 52. Cost, capacity

On screen:

> *Part V*
>
> *Where the bill lands, where the bottleneck moves, and what to do next week.*

Part five, the shortest and most concrete. [speculative] Three things: what a loop closed on an external check costs to run, where the constraint moves once you adopt it, and one recommendation for each of the three jobs in this room. [speculative] I kept the cost material to the end deliberately: a cost slide before an architecture argument turns into a procurement conversation. [speculative] But no CTO should leave without numbers, and no engineer should propose an architecture without knowing what it bills. [speculative]

53 53. The bill for closing a loop on an external check: more agents, more iterations, a memory layer

On screen:

> *Part V · Cost*
>
> *Several agents, not one*
>
> *A shared-transcript design: "approximately 15× more tokens than chats" [V38]. On the same model, caching and capped-summary sub-agent contracts "cut tokens ~38%" [V58]*
>
> *More loop iterations*
>
> *Reflexion, a self-reflection loop, "can become worse with more compute budget" [V39]*
>
> *A memory layer*
>
> *Pays back between turn 0 and turn 356; "no system wins on both cost and accuracy" [V40]*
>
> *Ask for this design's measured token multiplier, and memory's break-even turn. [speculative]*
>
> *[V38] [V58] [V39] [V40] Agentic Orchestration; Reflexion; Agent Memory.*

First: a shared-transcript multi-agent design costs roughly fifteen times the tokens of a straight chat. [V38] Say shared-transcript, because that is the architecture the benchmark measured and it is what produces the number — every subagent carrying the whole conversation. [V58] It is not a fixed property of orchestration: the same note reports caching, compaction and capped-summary sub-agent contracts cutting tokens about thirty-eight percent and cost about forty-one on the same model, so the lever is the design rather than the model version. [V58] Budget the fifteen times, then go and find your thirty-eight. [inference] Second, the counter-intuitive one: iteration is not monotone. [speculative] Unlike self-consistency, a reflection loop can get worse with more compute budget — spending more can lower accuracy. [V39] Third, memory layers, which every team adds by reflex: the break-even against simply resending the transcript ranges from turn zero to turn three hundred and fifty-six depending on the system, and no system wins on both cost and accuracy. [V40] So memory is a conditional optimisation with a precondition to check rather than assume. [inference]

54 54. A 25× input-price gap makes the tier you run the check on a design decision

On screen:

> *Part V · Cost*
>
> *Route the check down a tier, never the hard generation: checking is the smaller, more structured task. [inference]*
>
> - *Reported: a "25x input-price gap" between top-tier and cheap models [V49].*
> - *Its limit, scoped to legacy: "cheap-model routing to hard legacy tasks (already the worst-case for agents) amplifies the probability of silent failures and verification debt" [V49].*
> - *Verification level 1 again: a compiler run costs no tokens. [inference]*
>
> *[V49] The AI Cost Reckoning — vault watch note (2026).*

There is roughly a twenty-five times gap in input pricing between top-tier and cheap models. [V49] Which gives you a lever most teams are not pulling: run the expensive model to generate, and run a cheap model or, better, a plain program to check. [inference] The checking step is usually the smaller, more structured task, and it is exactly the one you can push down a tier or out of the model entirely. [inference] And the price: the same source says cheap-model routing to hard legacy tasks — already the worst case for agents, in its own parenthesis — amplifies the probability of silent failures and verification debt. [V49] That is legacy code; it says nothing about hard greenfield work. [inference] Which is one more argument for verification level one: a compiler run has no token cost at all. [inference]

55 55. Forbes reports Microsoft moving one division off Claude Code

On screen:

> *Part V · Cost*
>
> *Model price has become a purchasing decision, not just an engineering one. [inference]*
>
> - *Internal Claude Code licences end for the Experiences+Devices division by 30 June 2026; those engineers move to GitHub Copilot CLI [S18].*
> - *The report attributes it to cost and to compliance tooling — Purview DLP at the tooling layer — not to capability [S18].*
> - *One outlet, one division, reporting rather than a Microsoft announcement: token cost reaching procurement, not how enterprises generally pick an agent. [inference]*
>
> *Spend the cheap tier on the check, not the generation. [inference]*

> *[S18] Microsoft Ends Claude Code Licenses — Markman, Forbes (2026). https://www.forbes.com/sites/jonmarkman/2026/06/01/microsoft-ends-claude-code-licenses-as-it-pushes-copilot-cli/*

One enterprise instance in public reporting, and I want you to hear both halves of it. [speculative] Forbes reported on the first of June that Microsoft is ending internal Claude Code licences across its Experiences and Devices division — Windows, Microsoft 365, Outlook, Teams, Surface — by the thirtieth of June, and moving those engineers to GitHub Copilot CLI. [S18] Read the attribution carefully, because this is not purely a price story: the report names cost and it names compliance tooling — Microsoft wants Purview data-loss prevention wired in at the tooling layer, which Copilot CLI carries — and it says the decision was not a capability judgement between the two tools. [S18] Now the limits, because they matter more than the headline. One outlet, one division, and a journalist's reporting rather than an announcement from Microsoft. [S18] So take it as evidence that token cost has reached procurement, not as evidence about how enterprises in general choose a coding agent. [inference] What it changes for you is where the expensive tier goes: if the bill is now a purchasing conversation, the cheapest thing to push down a tier is the check, not the hard generation. [inference]

56 56. Generation got cheap; review and integration did not

On screen:

> *Part V · Capacity*
>
> *"AI-written code still needs to be reviewed, integrated, tested, and released by humans" [V52]*
>
> *+180% / +30%*
>
> *Coding activity vs releases shipped, 100k+ developers [V52]*
>
> *+15% / −7%*
>
> *Feature-branch vs main-branch throughput, CircleCI, 28.7M workflows [V53]*
>
> - *One company: 76% more pull requests once agents ramped — "organizational absorption became the constraint" [V41].*
> - *Correction: DORA — Google's DevOps delivery study — reversed its 2024 throughput penalty in 2025; only a correlational link to stability survived. [V51] [V42]*
>
> *[V41] Harness Engineering — CI · [V42] [V51] [V52] [V53] DORA Metrics.*

Start with the cleanest measurement: it observes behaviour rather than asking people. [speculative] Across more than a hundred thousand GitHub developers and three generations of tooling, coding activity rose about a hundred and eighty percent and commit-level output roughly tripled — and the same developers

shipped only about thirty percent more releases. [V52] The authors give the mechanism in one sentence: AI-written code still needs to be reviewed, integrated, tested, and released by humans. [V52] Generation got cheap; understanding, validating, integrating and owning the change did not. [inference] Same shape inside the pipeline: CircleCI telemetry over twenty-eight point seven million workflows shows feature-branch throughput up fifteen percent while main-branch throughput fell seven percent — acceleration where work is generated, contraction where it has to be integrated. [V53] Third, one company's own account: seventy-six percent more pull requests requiring review once agents ramped, and organisational absorption became the constraint. [V41] Now a correction I owe you. [speculative] DORA 2024 reported a throughput penalty from AI adoption; the 2025 report reversed it, and only the negative relationship with delivery stability persisted, so do not repeat the throughput number. [V51] Treat the rest as correlational: DORA is a repeated cross-section rather than a panel, and the verification-bottleneck framing quoted alongside it comes from a vendor with an interest in that conclusion. [V42]

57 57. Capture traces first, write the evals those traces require, then fail the merge on them

On screen:

> *Part V · Practice*
>
> *A trace is one run recorded end to end — input, tool calls, output, tokens. An eval is an assertion over that output. [V43]*
>
> *1*
>
> *Trace before evals*
>
> *"capture traces with cost metadata before writing a single custom eval" [V43]*
>
> *2*
>
> *Evals from traces*
>
> *Write the evals those traces require. [V43]*
>
> *3*
>
> *Fail the merge*
>
> *A failing eval blocks the merge; a dashboard does not. [V43] [inference]*
>
> *[V43] Harness Engineering — Evaluation and Observability Loops — vault report.*

One: instrument first. [speculative] Capture traces with cost metadata before you write a single custom evaluation, because the traces tell you which evaluations need to exist — and the ones you would have guessed are usually not those. [V43] Two: then write those evaluations, and run them identically in development and in production, so a regression means the same thing in both. [V43] Three: put the result on a gate. A failing evaluation blocks the merge; a dashboard only records that it failed, and only if somebody is looking at it. [V43] [inference] And the reason your existing monitoring will not cover this: a confidently wrong answer that passed every infrastructure check is invisible at the infrastructure layer. [V43] Latency was fine. [inference] Cost was fine. [inference] The answer was wrong.

58 58. If you write code — review the diff in a fresh session, gate the irreversible actions

On screen:

> *Part V · Monday*
>
> *Four changes that are configuration rather than architecture, and three things to say no to. [speculative]*
>
> ***Adopt this week***
>
> - *Ask your coding assistant to review a diff in a fresh session, not the one that wrote it. [V17]*
> - *Isolate your checker's context — artifact and criteria only, never the transcript. [V17]*
> - *Put irreversible actions behind a deterministic gate. [V20]*
> - *Turn on tracing with cost metadata before writing evals. [V43]*
>
> ***Refuse this week***
>
> - *A second model reading the first one's output, labelled verification. [V3]*
> - *A twenty-step chain with one check at the end. [V27]*
> - *An agent credential scoped wider than its one task. [S9]*

Adopt, and the first needs no agentic system [speculative]: run your assistant's review of a diff in a fresh session, pasting the diff and the criteria only — a checker seeing the thread that wrote the code is biased toward confirming it. [V17] Same rule if you run a loop. [V17] Put irreversible actions behind a deterministic gate; the list of those actions is shorter than you think. [V20] Turn on tracing with cost metadata now, before you write any evaluations, because the traces will tell you which ones to write. [V43] Refuse: a second model reading the first model's output described as verification, because that is

generation wearing a different prompt. [V3] Refuse a twenty-step chain with a single check at the end — put the check on the seam that has the worst recovery rate. [V27] And refuse any agent credential scoped wider than the single task it was issued for. [S9]

59 59. If you judge output — ask which of the five levels checked it, and who chose the source

On screen:

> *Part V · Monday*
>
> *1*
>
> *Which level checked it?*
>
> *Level 4 — a model scoring a rubric — is coverage, not certification. [V18]*
>
> *2*
>
> *Faithful, or true?*
>
> *Ask who selected the source, and what verified that selection. [V15]*
>
> *3*
>
> *How often, not how well?*
>
> *Ask for all-k-succeed, not best-of-k. Demand the reliability floor. [V44]*
>
> *Verified means you can name what checked it. These three questions get that name, without a compiler. [inference]*
>
> *[V15] [V18] Loop Engineering · [V44] Unit Testing.*

Three questions, none of which requires you to read a model card. [speculative] One: which of the five verification levels verified this decision — and if the answer is a model scored it, that is level four, which is coverage and cannot carry a gate. [V18] Two: was the answer faithful, or was it true? [speculative] Ask who selected the source document and what verified that selection, because a perfectly faithful quotation of the wrong source scores above ninety percent on the metric you will be shown. [V15] Three: how often rather than how well — ask for the rate at which all attempts succeed, not the best of several attempts. [V44] Three questions, no compiler required.

60 60. If you decide budgets — fund the oracle, fund review capacity, budget the re-scope

On screen:

> *Part V · Monday*
>
> *1*
>
> *Fund the oracle*
>
> *Formalising domain knowledge costs engineering time a generic vendor cannot copy. [S5] [V26]*
>
> *2*
>
> *Fund review capacity*
>
> *Review capacity is the constraint that showed up first in practice. [V41] [V52]*
>
> *3*
>
> *Budget the depreciation*
>
> *Re-scope every model-judge step at each model upgrade. [V25] [S2]*
>
> *Set the autonomy level per feature, with a named person against each. [V34] And fund on the rate at which all attempts succeed, per stage — not an end-to-end average. [V44] [V29]*

Three line items nobody puts in the plan. [speculative] One: fund the oracle. [speculative] Formalising your domain knowledge into machine-checkable form is the expensive part, and it is also the part a generic vendor cannot copy — the verifier may be more work than the generator, and someone has to write it. [S5] [V26] Two: fund absorption. [speculative] Review capacity was the constraint that appeared first in every account I could find, and buying more generation without buying more absorption converts directly into review debt. [V41] [V52] Three: budget the depreciation — every model-judge step needs re-scoping at each model upgrade, so put it in the maintenance plan rather than discovering it in the bill. [V25] [S2] And set the autonomy level explicitly per feature, with a named person against each, because that decision is being made either way. [V34] And the bar for reliable: the rate at which all attempts succeed, per stage — an end-to-end average hides the stage that recovered zero percent. [V44] [V29]

61 61. What I do not know — five limits; if a system you run rests on one, say so in the design review

On screen:

> *Honest limits*
>
> •
>
> *Retrofit*
>
> *[V22]*
>
> *"There is no convincing public playbook yet for retrofitting a ten-year-old codebase."*
>
> •
>
> *Oracle coverage*
>
> *[V6]*
>
> *No general theory of which failures an oracle cannot see.*
>
> •
>
> *Loop drift*
>
> *[V19]*
>
> *No controlled study of single-loop degradation over many cycles.*
>
> •
>
> *Self-critique bound*
>
> *[V13]*
>
> *The information-theoretic bound rules out no design until independently operationalised.*
>
> •
>
> *Adoption effects*
>
> *[V51]*
>
> *DORA reversed its 2024 throughput finding; the rest is correlational.*

Retrofit: there is no convincing public playbook for retrofitting a ten-year-old codebase, and every success story I can cite is greenfield — so if you are adding this to a system that already exists, you are past the published evidence. [V22] Oracle coverage: there is no general theory of which failures a given oracle cannot see; it is judgement, system by system, every time. [V6] Loop drift: nobody has run a controlled study of how a single unattended loop degrades over many cycles, so anyone running one for weeks is operating past the evidence. [V19] The self-critique bound: the information-theoretic bound needs independent operationalisation, so it rules out no design. [V13] And the adoption figures are unstable as well as correlational — DORA reversed its own throughput finding between the 2024 and 2025 reports, so

anything I tell you about adoption effects on delivery has roughly a one-year half-life. [V51] If a system you run rests on any of these five, that belongs in the design review as an open question, not as a citation to me. [speculative]

62 62. Takeaways — move the check outside the model, and authority out of the prompt

On screen:

> *Where this went*
>
> *1*
>
> *Externalise the check*
>
> *Structural code compliance 56.8% → 98.6% — the external verifier, not a better model. [V5]*
>
> *2*
>
> *Isolate the checker*
>
> *Same model, fresh context: the artifact and the criteria, never the transcript. [V17]*
>
> *3*
>
> *Name the level*
>
> *Which of the five levels checked this? Level 4, a model scoring a rubric, is coverage, not certification. [V18]*
>
> *4*
>
> *Gate irreversible actions*
>
> *Irreversible actions on a deterministic gate; autonomy level named. [V20] [V34]*

One: move the truth-determining computation out — the study's own authors attribute the jump from fifty-six to ninety-eight percent to closing the loop on an external verifier rather than to a more capable model, and it held across backbone models. [V5] Two: isolate the checker — same model, fresh context, shown the artifact and the acceptance criteria and never the generator's transcript, because a checker that reads the reasoning is biased toward confirming it. [V17] Three: name the verification level every time, and treat a model-as-judge step as coverage, not certification. [V18] Four: add authority last — irreversible actions on a deterministic gate, and an autonomy level chosen deliberately with a name attached to it. [V20] [V34] If you remember one sentence: the agent stated 90 to 100 confidence and caught none of the thirty-four violations, so put the check outside the model. [V1]

63 63. What follows from an external check that can fail the work, and a deny gate at the credential

On screen:

> *Closing*
>
> *If reliability comes from those two mechanisms, four things follow. [inference]*
>
> - *The check is the hard part, not the prompt: unit tests, schema validation, compile steps over model output. [inference]*
> - *~16 points on a LongMemEval subset (116 of 500) came from the harness, not the model; Chronos is the authors' own harness [V23].*
> - *Verification has a price [V14]: a second pass, another model, a deterministic check, someone's time. Name who re-scopes it at upgrades. [speculative]*
> - *Name what fails when the agent is wrong: a red test, a rejected schema, a denied deploy credential. [inference]*
>
> *[V14] Loop Engineering · [V23] Is Grep All You Need?*

First, the hard part moved from the prompt to the check — and your team already writes checks: unit tests, schema validation, a compile-and-run step, now run over model output. [inference] Second, model choice matters less than harness and oracle design: the sixteen-point accuracy swing, on a hundred-and-sixteen-question subset, came from the wrapper, not the weights. [V23] Third, verification is always paid for, and you can name the currency: a second inference pass, another model, a deterministic check, or review time. [V14] So my bet is that the job that lasts is owning that check: re-scoping it at every model upgrade because a verifier depreciates [V25], and standing behind its verdicts. [speculative] A check nobody owns keeps passing work it stopped covering. [speculative] So the hour reduces to something you can do on Monday: take one agent you run and name the artifact that fails when it is wrong — the test that goes red, the schema that rejects, the credential that denies the deploy. [inference] For this deck those artifacts are the quote checker and my own read of every slide. [inference] If nothing on that list exists for your system, it is unverified however well its output reads. [inference] Thank you. [speculative] Questions.

64 64. References

On screen:

- ***[V1]*** *Loop Engineering and Self-Verification Loops — Practitioner Synthesis — vault research report (2026), on the 0-of-34 self-audit against a deterministic check.*
- ***[V3]*** *Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026), on the discriminator: a check that can come back negative.*
- ***[V4]*** *Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026), on structural code compliance, 56.8% to 98.6%.*
- ***[V5]*** *Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026), on the attribution to the external verifier, and the backbone swap.*
- ***[V6]*** *Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026), on the parameter-level loop being powerless against a topology defect.*
- ***[V7]*** *Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026), on clinical-diagnosis accuracy, 55.6% to 77.5%.*
- ***[V8]*** *Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026), on the proof-checker oracle and the 422% prover figure.*
- ***[V9]*** *Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026), on on-policy self-monitoring: 5× approval, AUROC 0.99 to 0.89.*
- ***[V10]*** *Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026), on generation capability not carrying self-verification.*
- ***[V11]*** *Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026), on the consistency dilemma.*
- ***[V12]*** *Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026), on the assumption that verification is easier than generation.*
- ***[V13]*** *Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026), on the self-critique bound as a diagnostic framework.*
- ***[V14]*** *Loop Engineering and Self-Verification Loops — Practitioner Synthesis — vault research report (2026), on verification always having a price.*
- ***[V15]*** *Loop Engineering — Practitioner Synthesis — vault report, on the faithfulness–truth gap.*
- ***[V16]*** *Loop Engineering and Self-Verification Loops — Practitioner Synthesis — vault research report (2026), on self-verification not improving with scale.*

- ***[V17]*** *Loop Engineering — vault concept note (2026), on checker independence as context isolation.*
- ***[V18]*** *Loop Engineering — vault concept note (2026), on the five verification levels, and 70% at levels 1–2.*
- ***[V19]*** *Loop Engineering — vault concept note (2026), on the hand-coded corpus of fifty production loops.*
- ***[V20]*** *Loop Engineering — vault concept note (2026), on irreversible actions belonging on a deterministic deny gate.*
- ***[V21]*** *Harness Engineering — vault concept note (2026), on the definition of a harness.*
- ***[V22]*** *Harness Engineering — vault concept note (2026), on the missing retrofit playbook.*
- ***[V23]*** *Is Grep All You Need? How Agent Harnesses Reshape Agentic Search — vault reading note (2026), on the ~16-point harness gap on a LongMemEval subset.*
- ***[V24]*** *Is Grep All You Need? How Agent Harnesses Reshape Agentic Search — vault reading note (2026), on grep 93.1% against vector 83.6%.*
- ***[V25]*** *2026-07-21 Loop Engineering — Verification and Checker Design — vault research report (2026).*
- ***[V26]*** *Structural Grounding — vault concept note (2026).*
- ***[V27]*** *Agentic Loop Pattern — vault Zettelkasten note (2026).*
- ***[V28]*** *Grounding in AI-Driven Systems — Composition, Agentic Patterns, End-to-End Guarantees — vault podcast brief (2026).*
- ***[V29]*** *The 4/δ Bound — Designing Predictable LLM-Verifier Systems — vault reading note (2026).*
- ***[V30]*** *PocketOS Cursor Incident — Database Deletion in Nine Seconds — vault watch note (2026), on the nine-second deletion and its sequence.*
- ***[V31]*** *PocketOS Cursor Incident — Database Deletion in Nine Seconds — vault watch note (2026), on system prompts not being security controls.*
- ***[V32]*** *Harness Engineering — Tool Permissions and Sandboxing — vault research report (2026).*
- ***[V33]*** *Guardrails and Runtime Enforcement — vault report.*
- ***[V34]*** *Bounded Autonomy in AI — vault concept note (2026).*
- ***[V35]*** *Agentic Coding Is a Trap — The Code newsletter (2026-05-15).*
- ***[V37]*** *2026-07-22 Digest — vault journal note (2026).*
- ***[V38]*** *Agentic Orchestration Patterns — vault concept note (2026).*
- ***[V39]*** *Reflexion Cost-Efficiency Tradeoffs, Failure Modes, Iteration Safety Bounds — vault concept note (2026).*
- ***[V40]*** *Harness Engineering — Agent Memory Architecture — vault research report (2026).*

- ***[V41]*** *Harness Engineering — Workflow and CI Orchestration — vault research report (2026).*
- ***[V42]*** *DORA Metrics Limitations in AI-Assisted Development — vault Zettelkasten note (2026).*
- ***[V43]*** *Harness Engineering — Evaluation and Observability Loops — vault research report (2026).*
- ***[V44]*** *Unit Testing — vault concept note (2026).*
- ***[V45]*** *Building AI Systems — predecessor-talk glossary — vault talk note (2026).*
- ***[V46]*** *Claude 5 self-verification and the orchestration premium — vault tutor card (2026-07-29).*
- ***[V47]*** *2026-07-18 Digest — vault journal note (2026).*
- ***[V48]*** *2026-07-17 Digest — vault journal note (2026).*
- ***[V49]*** *The AI Cost Reckoning — Right-Sizing Model Spend 2026 — vault watch note (2026).*
- ***[V50]*** *Context Engineering — vault Zettelkasten note (2026).*
- ***[S1]*** *Building Effective Agents — Anthropic. https://www.anthropic.com/engineering/building-effective-agents*
- ***[S2]*** *Harness design for long-running application development — Anthropic. https://www.anthropic.com/engineering/harness-design-long-running-apps*
- ***[S3]*** *Large Language Models Cannot Self-Correct Reasoning Yet — Huang et al. (ICLR 2024). https://arxiv.org/abs/2310.01798*
- ***[S4]*** *On the Self-Verification Limitations of LLMs on Reasoning and Planning Tasks — Stechly, Valmeekam, Kambhampati (ICLR 2025). https://arxiv.org/abs/2402.08115*
- ***[S5]*** *LLMs Can't Plan, But Can Help Planning in LLM-Modulo Frameworks — Kambhampati et al. https://arxiv.org/html/2402.01817v2*
- ***[S6]*** *Towards a Science of AI Agent Reliability — Rabanser, Kapoor et al. (ICML 2026). https://arxiv.org/abs/2602.16666*
- ***[S7]*** *Verification-Driven Closed-Loop Multi-Agent LLM Framework for Code-Compliant Structural Design — arXiv (2026). https://arxiv.org/abs/2608.07978*
- ***[S8]*** *Ironies of Automation — Bainbridge, Automatica (1983). https://ckrybus.com/static/papers/Bainbridge_1983_Automatica.pdf*
- ***[S9]*** *The Protection of Information in Computer Systems — Saltzer and Schroeder (1975). https://www.cs.virginia.edu/~evans/cs551/saltzer*
- ***[S10]*** *Self-Correction as Feedback Control: Error Dynamics, Stability Thresholds, and Prompt Interventions in LLMs — arXiv preprint (2026). https://arxiv.org/html/2604.22273v2*

- ***[S11]*** *Concepts — tokens — OpenAI documentation. https://platform.openai.com/docs/concepts*
- ***[S12]*** *AI Bias May Impair Radiologist Accuracy — RSNA (2023). https://www.rsna.org/news/2023/may/ai-bias-may-impair-accuracy*
- ***[S13]*** *Decomposing LLM Self-Correction: The Accuracy-Correction Paradox and Error Depth Hypothesis — arXiv preprint (2026). https://arxiv.org/abs/2601.00828*
- ***[V51]*** *DORA Metrics Limitations in AI-Assisted Development — vault Zettelkasten note (2026), on the 2025 throughput reversal.*
- ***[V52]*** *DORA Metrics Limitations in AI-Assisted Development — vault note, citing Demirer et al. (2026), 100,000+ GitHub developers.*
- ***[V53]*** *DORA Metrics Limitations in AI-Assisted Development — vault note, citing CircleCI, State of Software Delivery (2026).*
- ***[V54]*** *Loop Engineering — vault concept note (2026), on the ceiling of checker independence.*
- ***[V55]*** *Anthropic — How We Contain Claude Across Products — vault watch note (2026-05-27), summarising https://www.anthropic.com/engineering/how-we-contain-claude*
- ***[V56]*** *Harness Engineering — vault concept note (2026), on the Terminal-Bench 2.1 model-versus-harness spread.*
- ***[V57]*** *Harness Engineering — Guardrails and Runtime Enforcement — vault research report (2026), on indirect prompt injection.*
- ***[V58]*** *Agentic Orchestration Patterns — vault concept note (2026), on the architecture behind the 15× token multiplier.*
- ***[S14]*** *AI Agent Orchestration Patterns — Microsoft Azure Architecture Center (2026). https://learn.microsoft.com/en-us/azure/architecture/ai-ml/guide/ai-agent-design-patterns*
- ***[S15]*** *Tool Calling — NVIDIA NeMo Guardrails Library Developer Guide — NVIDIA (2026). https://docs.nvidia.com/nemo/guardrails/configure-guardrails/guardrail-catalog/tool-calling*
- ***[S16]*** *AI Act Annex III HR & Employment: High-Risk AI Compliance Guide for 2026 — Knowlee (2026). https://www.knowlee.ai/blog/ai-act-annex-iii-hr-employment*
- ***[S17]*** *Regulation (EU) 2024/1689 (EU AI Act) — Article 14, Human Oversight — European Union (2024). https://artificialintelligenceact.eu/article/14/*
- ***[S18]*** *Microsoft Ends Claude Code Licenses As It Pushes Copilot CLI — Jon Markman, Forbes (2026). https://www.forbes.com/sites/jonmarkman/2026/06/01/microsoft-ends-claude-code-licenses-as-it-pushes-copilot-cli/*

The handout PDF carries the full reference list, and a one-page summary of every [V] source with its own external references.

Everything on a slide tonight traces to one of these seventy-two entries, and the numbering matches the handout. [speculative] Two prefixes: S is a published source you can open from the URL beside it, V is a note from my own vault — and the handout summarises each V note with every external source it draws on. [speculative] Is Grep All You Need is the harness-versus-model measurement I leaned on twice — one 2026 study, so read it before repeating my sixteen-point figure. [speculative] Read the PocketOS post-mortem in full: the sequence of small permission decisions teaches more than the headline. [speculative] Three entries make human approval uncomfortable — skill atrophy, the Article 14 text, the liability framing. [speculative] The DORA note I corrected on the slide rather than dropped, and the Demirer and CircleCI entries beside it are what replaced the retired throughput number. [speculative] Huang and Stechly are the two negative results — read their scope sections, not the abstracts: Huang is specifically about intrinsic self-correction on reasoning tasks, and that scope is what people drop. [speculative] On Stechly, I quote the per-domain figures from published summaries, not from the PDF. [speculative] The LLM-Modulo paper rejects both extremes, and it is what made me soften my thesis. [speculative] The structural-design preprint is the primary source for my headline number, in a domain that is not yours. [speculative] Bainbridge is 1983, Saltzer and Schroeder 1975, which is the point. [speculative] The two self-correction preprints make it conditional rather than forbidden; the second is one maths benchmark and three models, so read its ratio as a measurement, not a curve. [speculative] How We Contain Claude is a vendor writing about its own sandbox, and the leaderboard snapshot will move. [speculative]